

Canonical 12-D Spatio-Temporal Feature Projection for Zero-Shot Cross-Network Transfer

Raw Heterogeneous Financial Schemas

Bitcoin UTXO (DAG)
166-D Raw Attributes

Ethereum EVM (Calls)
14-D Smart Contracts

PaySim (Mobile Money)
9-D High-Velocity Tree

SAML-D (Multi-Bank)
12-D Wire Mesh

12-D Canonical Invariant Projection Space ($\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$)

Physical Mass-Conservation & Causal Graph Topological Invariants

1. Flow Conservation: $\Phi_{\text{flow}}(u, t) = \Sigma A_{\text{out}} / (\Sigma A_{\text{in}} + \epsilon)$
→ *Layering Mule Flow Balance (~1.0 unity ratio)*
2. Degree Asymmetry: $\Psi_{\text{asym}}(u, t) = |d_{\text{in}} - d_{\text{out}}| / (d_{\text{in}} + d_{\text{out}} + \epsilon)$
→ *Conduit vs High-Degree Structural In/Out Skew*
3. Hawkes Arrival: $\lambda_u(t) = \mu_0 + \Sigma \alpha e^{-\beta(t-t_i)}$
→ *Continuous-Time Micro-Burst Velocity Acceleration*
- 4-5. PPR Taint: $(\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}) = (\mathbf{I} - \alpha_{\text{ppr}} \mathbf{P})^{-1} \mathbf{s}_{\text{seed}}$
→ *Causal Illicit Taint Network Proximity (Bi-Directional)*
- 6-12. 5-Moment & Jitter: $(\mu_A, \sigma_A^2, \gamma_A, \kappa_A, \nu_{\text{io}}, \sigma_{\Delta t}, \nu_A)$
→ *Higher-Order Ego Distribution Statistics & Temporal Dispersion*

Frozen GNN Backbone & Target Transfer

100% Frozen Weights
Zero target fine-tuning ($\eta = 0$)

Universal Alignment
Cross-rail canonical basis

Target Transfer F1
82.40% Macro F1 Score

Invariant Advantage
+28.20 pp vs Raw Projection